🧩 STEP 1: Install Git in Ubuntu

Open Terminal and run:

```
sudo apt update
sudo apt install git -y
```

Check:

```
git --version
```

✓ If version shows → Git installed successfully

🧩 STEP 2: Configure Git

```
git config --global user.name "Your Name"
git config --global user.email "your@email.com"
```

Check config:

```
git config --list
```

🧩 STEP 3: Create Project Folder

```
mkdir data-science-project
cd data-science-project
```

🧩 STEP 4: Initialize Git Repo

```
git init
```

🧩 STEP 5: Connect to GitHub

Go to GitHub → create repo
Then run:

```
git remote add origin https://github.com/your-username/data-science-project.git
git branch -M main
```

🧩 STEP 6: Create program1.py

```
nano program1.py
```

Paste:

```
num = int(input("Enter a number: "))

print("Multiplication Table of", num)
```

```
for i in range(1, 11):
    print(num, "x", i, "=", num * i)
```

👉 Save:

- Press CTRL + X
- Press Y
- Press ENTER

🧩 STEP 7: Commit & Push

```
git add .
git commit -m "Added program1"
git push -u origin main
```

⚠️ IMPORTANT (GitHub Authentication)

If it asks password ❌
👉 Use **Personal Access Token (PAT)** instead

Steps:

1. Go to GitHub → Settings
2. Developer Settings → Tokens
3. Generate token
4. Use token instead of password

🧩 STEP 8: Create Branch

```
git checkout -b GIT-EXAM
```

🧩 STEP 9: Create program2.py

```
nano program2.py
```

Paste:

```
a = int(input("Enter first number: "))
b = int(input("Enter second number: "))

if a > b:
    print("Greatest number is:", a)
else:
    print("Greatest number is:", b)
```

Save (same as before)

🧩 **STEP 10: Commit & Push**

```
git add .
git commit -m "Added program2"
git push -u origin GIT-EXAM
```

---

open project file in git dextop the run

🧩 **STEP 11: Show Diff**

```
git diff main GIT-EXAM
```

👉 SAME command works ✓

QNA

🧪 🔥 **WHAT THIS PRACTICAL IS ABOUT (Simple Meaning)**

👉 This practical is about using **Git** and **GitHub** to:

- Track your code changes
- Store code online
- Work with different versions (branches)
- Compare changes

🧠 **WHY WE DO THIS PRACTICAL**

👉 Imagine this real situation:

You are building a project:

- You write code today
- Tomorrow you make changes
- Suddenly something breaks ❌

👉 Without Git:

- You lose old code

- No backup
- No tracking

👉 With Git:

- Every change is saved ✓
- You can go back anytime ✓
- You can compare versions ✓

⭐ **IMPORTANCE (VERY IMPORTANT FOR VIVA)**

1. 🔄 Version Control

👉 Tracks every change in code
👉 Like "history" of your project

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2. 🌐 Backup on Cloud

👉 Using GitHub
👉 Your code is safe online

---

3. 🌿 Branching (MOST IMPORTANT)

👉 You created GIT-EXAM branch

👉 Why?

- To work on new feature without affecting main code
- Safe experimentation

---

4. 👨‍💻 Team Collaboration

👉 Many developers can work together
👉 No code conflict

---

5. ⚙️ DevOps Foundation

👉 Git is base for:

- CI/CD
- Automation
- Deployment

🔍 **WHAT YOU DID IN PRACTICAL (Explain Like This)**

👉 Say this:

"First, we initialized a Git repository. Then we created a Python file and pushed it to GitHub. After that, we created a new branch to add another program, committed the changes, and used git diff to compare both branches."

🧠 **CORE CONCEPTS (VERY IMPORTANT)**

🔹 **1. Repository (Repo)**

👉 Folder where your project is stored with Git tracking

🔹 **2. Commit**

👉 Saving changes with message
👉 Like checkpoint

🔹 **3. Branch**

👉 Separate version of code

Example:

- main → original
- GIT-EXAM → new feature

🔹 **4. Push**

👉 Upload code to GitHub

🔹 **5. Pull**

👉 Download latest code

🔹 **6. Diff**

👉 Shows difference between versions

🎯 MODERATE LEVEL VIVA QUESTIONS + ANSWERS

❓ What is Git?

👉
Git is a distributed version control system used to track changes in code and manage different versions of a project.

❓ What is GitHub?

👉
GitHub is a cloud platform that hosts Git repositories and allows collaboration and code sharing.

❓ Difference between Git and GitHub?

👉

- Git → tool (local)
- GitHub → platform (online)

❓ What is a branch?

👉
A branch is an independent version of the code used to develop features without affecting the main project.

❓ Why did you create GIT-EXAM branch?

👉
To add a new program without modifying the main branch, ensuring safe development.

❓ What is commit?

👉
A commit is a snapshot of changes saved in the repository.

❓ What is git diff?

👉
It shows the difference between two branches or commits.

❓ **What is origin?**

👉
Origin is the default name for the remote GitHub repository.

❓ **What is HEAD?**

👉
HEAD points to the current branch or latest commit.

❓ **What happens if two people edit same file?**

👉
Git detects conflict and asks to resolve it manually.

🔥 **SLIGHTLY ADVANCED (FOR IMPRESSING EXAMINER)**

❓ **What is distributed version control?**

👉
In Git, every user has a full copy of the repository, not just a central server.

❓ **What is staging area?**

👉
It is an intermediate area where changes are prepared before committing.

❓ **What is CI/CD?**

👉

- CI → Continuous Integration
- CD → Continuous Deployment

👉 Automates build and deployment

🧠 **ONE-LINE POWER ANSWER**

👉
"This practical demonstrates version control using Git and GitHub, including branching, committing, and comparing changes, which are essential for collaborative and efficient software development."